Data Source

From Amap

A. Route plans

Multiple detailed routes and associated ODs per plan

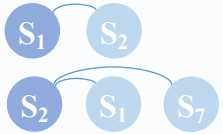
13M route plans

B1. Station Information

120K stations

B2. Station Connectivity

First-order station connectivity



…

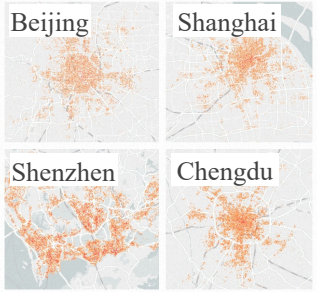
C. Line Information

Line length, stop sequences, operating hours

…

13K transit lines

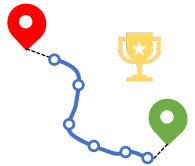
Covers 4 cities



TransitBench

3 Evaluation Tasks

Optimal Route Generation (ORG)



10K

Personalized Route Generation (PRG)

10K

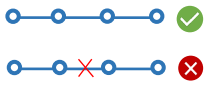
Diverse Route Generation (DRG)

10K

10 Evaluation Metrics Across 5 Categories

1. Connectivity

Conn



2. Access Feasibility

Station Grounding
SG

Distance Plausibility
DP
Accuracy of feeder distance estimation

3. Route Overlap

Consistency with ground truth routes

Line Overlap
LO

Station Sequence Overlap
SSO

Route Exact Match
REM
LO = 1 & SSO = 1

4. Numeric Field Accuracy

Correctness of predicted route features

Estimation Accuracy
EA

Mean Absolute Percentage Error
MAPE

5. Task-Specific

Preference Compliance
PC

Route Diversity
RD

TransitLM

General LLMs perform poorly on TransitBench

…

Our Solution: TransitLM

TransitData

Exclude benchmark data from the data sources

CPT

3 data sources → Align → Text

	Knowledge	Ratio
A	route	Various
B1	station	100%
B2	connectivity	100%
C	line	100%

SFT

3 Tasks

ORG PRG DRG

30k samples for training

Input Formats

Prompt: task definition

Input: query + OD information

Training

Varied Data Settings

Multi-CPT × Multi-SFT

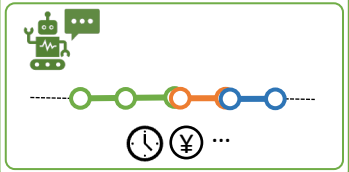
Vocabulary Expansion

Stations → IDs

3 Base Model Sizes

0.6B / 1.7B / 4B

Evaluation

On TransitBench

Query = [O→D] + {Preference, k-routes}*